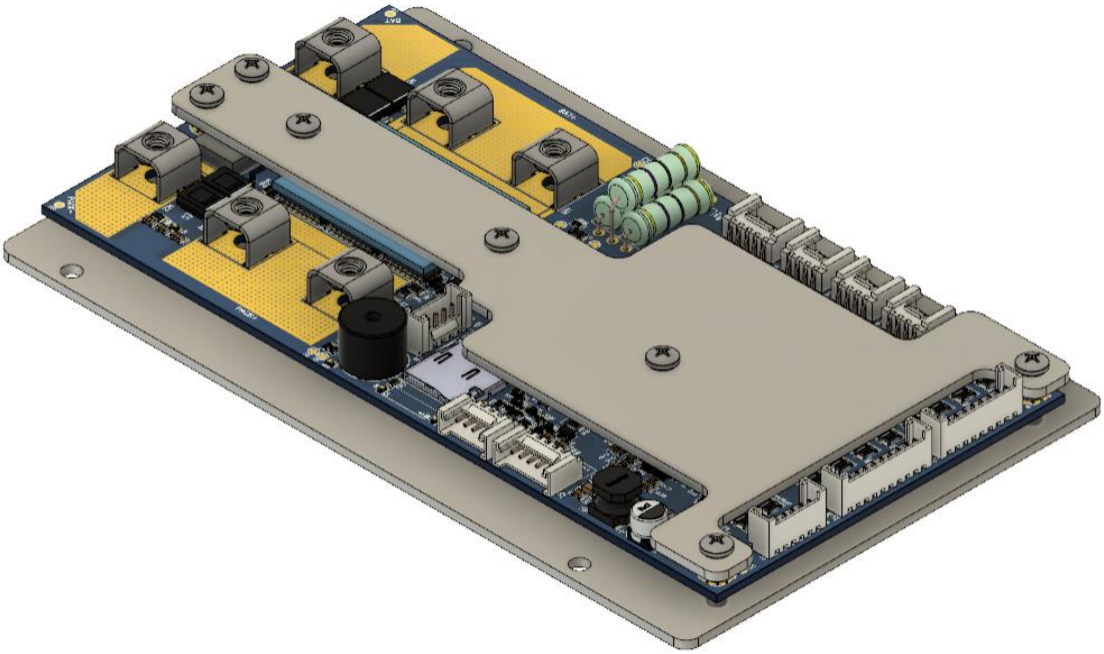
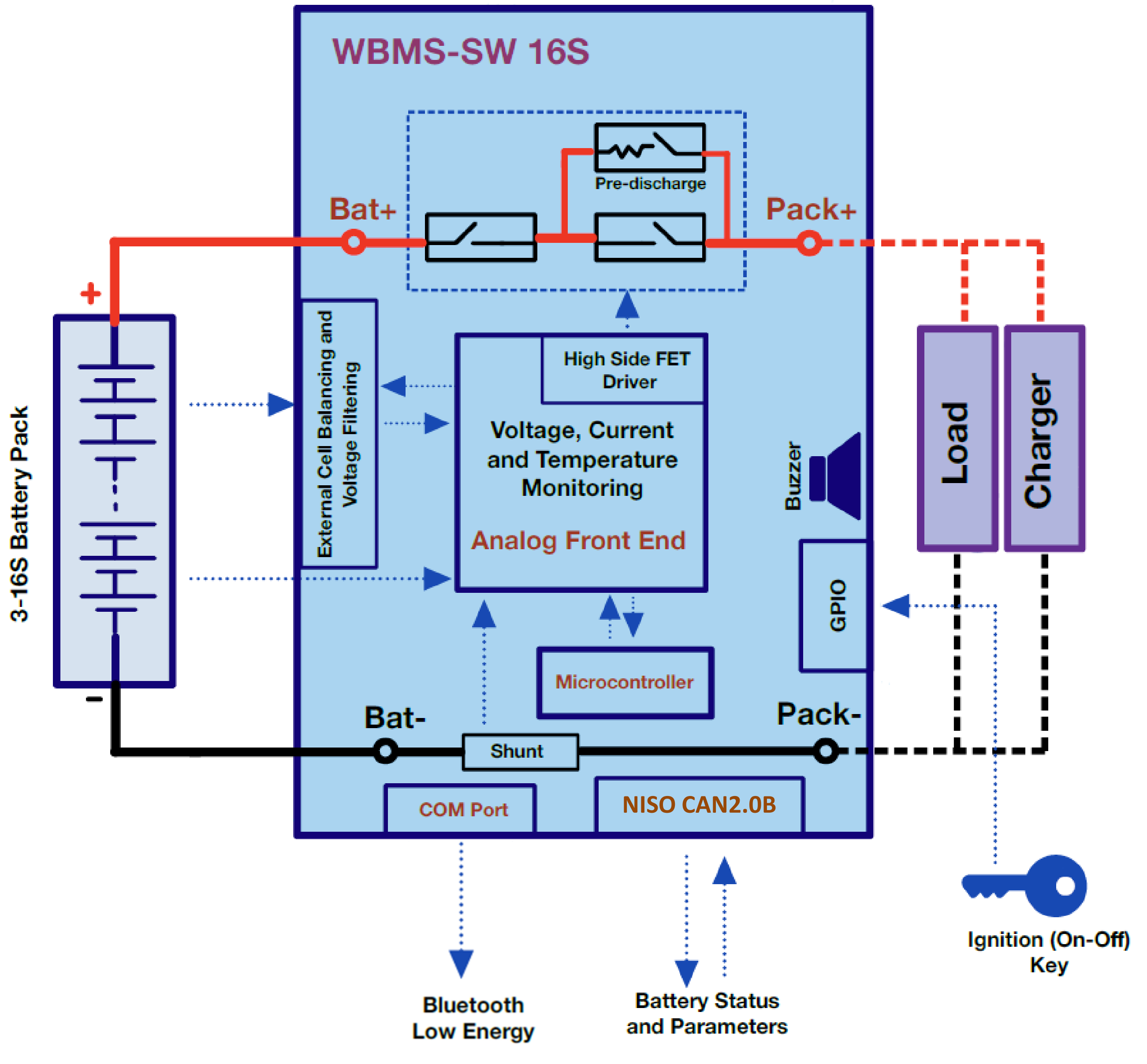
WBMS-SWLT 20S 60A

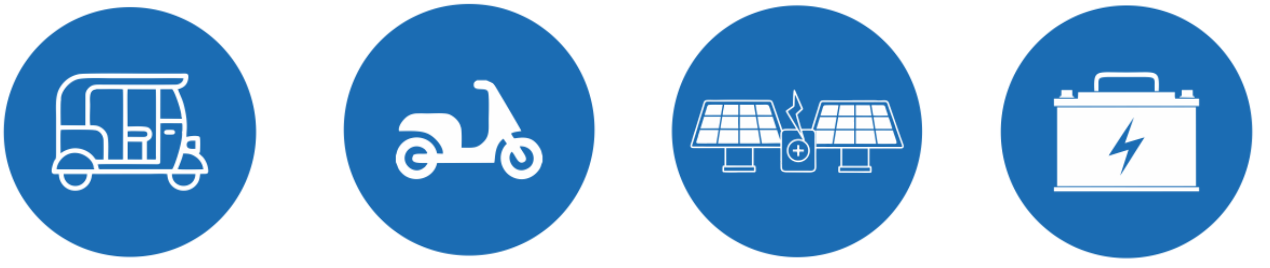
Smart BMS Manages Lithium-Ion Batteries with Precision and Speed

Specification Sheet v1.0



Application Diagram





Salient Features

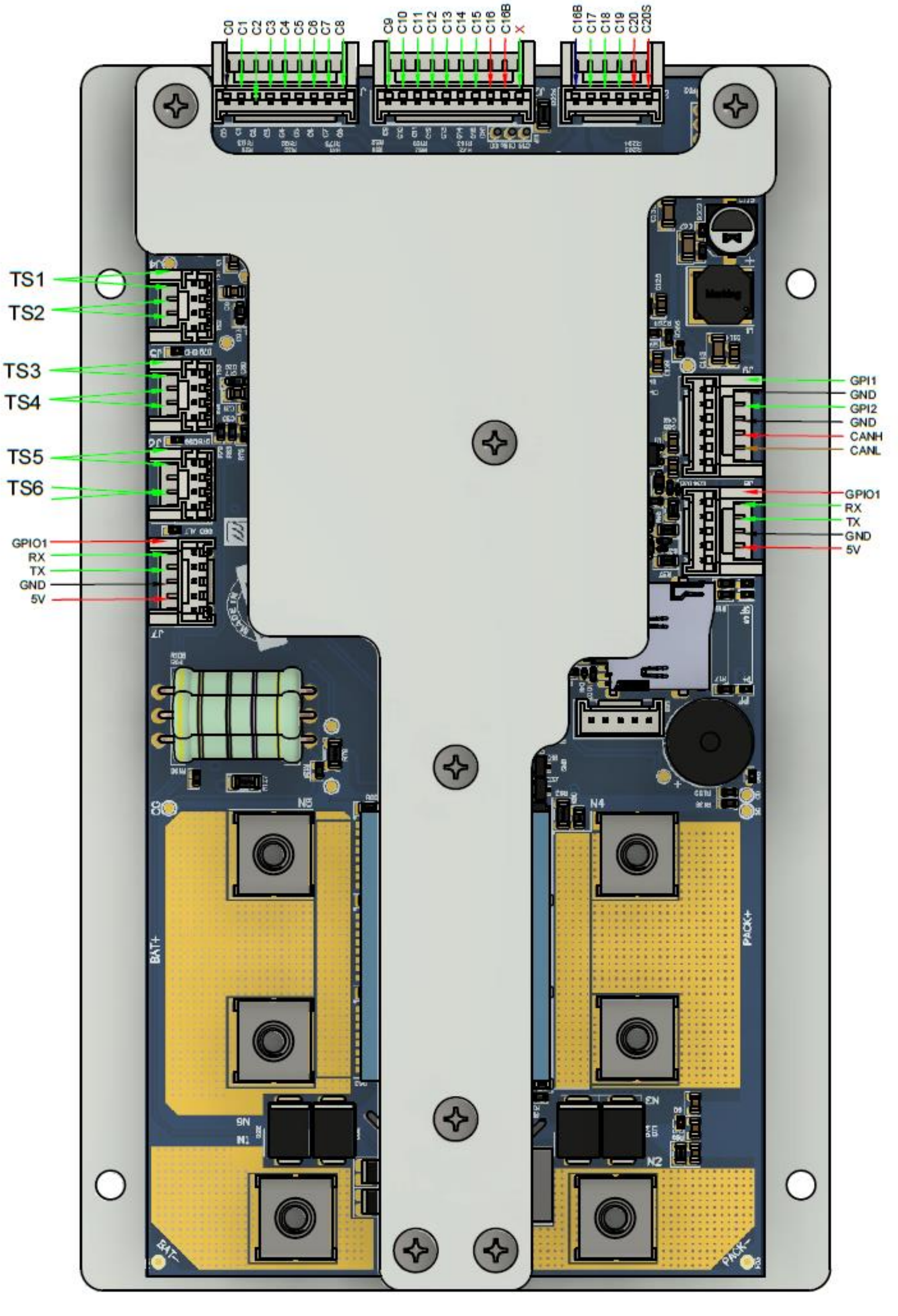
- Scalable up to **17S to 20 cells** & **48-73V** battery packs
- **AIS-156 PH-II Compliant**
- **Buzzer output** and **LED Indication** for **Thermal Runaway, Open Wire Fault** and **MOSFET Failure**
- Multiple communication options: **CAN2.0** and **BLE**
- Onboard: **uSD Card** or **EEPROM**
- **High side N-MOSFET control** provides safe and uninterrupted ground connection to external subsystems.
- **Predischarge control** for preventing inrush currents to capacitive loads. Prevents MOSFET failures and increases cell life
- **Short Circuit and Overcurrent in discharge latches** ensure automatic locking of discharge path after repeated short circuit and overcurrent faults
- **Cell balancing current up to 160mA** minimizes cell to cell voltage and SOC variations and optimal charging of battery pack
- Individual Cell over and under voltage and temperature protection
- Pack level over current and short circuit protection
- **Deep Discharge Prevention (Auto BMS Shutdown)** when cell voltage is critically low
- Proprietary **CAN based field servicing tools** for **data monitoring, logging** and **firmware update**

<table>
<tr><th colspan="2">Specifications</th><th colspan="2">Remarks</th></tr>
<tr><td colspan="2">Number of Cells</td><td colspan="2">Min: 17 Max: 20</td></tr>
<tr><td colspan="2">Cell Voltage Sensing Range</td><td colspan="2">1.6 – 3.7V</td></tr>
<tr><td colspan="2">Peak Battery Terminal Voltage</td><td colspan="2">73 V (20S)</td></tr>
<tr><td colspan="2">Sensing Accuracy</td><td colspan="2">Cell Voltage: ±3mV Current: ±1%</td></tr>
<tr><td colspan="2">Dimension</td><td colspan="2">171 mm X 103 mm X 16 mm (L X B X H)</td></tr>
<tr><td>Rated Nominal Continuous Current Rating</td><td>Charging and Discharging Current</td><td colspan="2">Up to 100A. Needs thermal connection to metal battery pack casing for PDU heat dissipation to maintain continuous current rating. Continuous current rating to be derated to ~65% of the rated value if no thermal connection is present.</td></tr>
<tr><td>Cell Balancing</td><td>Balancing Current per Cell</td><td colspan="2">100~160 mA</td></tr>
<tr><td>Pre-discharge control</td><td>Nominal Capacitive load</td><td colspan="2">3-6mF in 0.5 seconds</td></tr>
<tr><td>BMS Self Current Consumption</td><td>Active Mode, Shutdown Mode</td><td colspan="2">Active Mode: ~13mA
Shutdown Mode: ~0.15mA</td></tr>
<tr><td rowspan="3">Temperature Sensing</td><td>Onboard</td><td colspan="2">One channel for onboard MOSFET temperature</td></tr>
<tr><td>External</td><td colspan="2">Six external channels for cell temperature</td></tr>
<tr><td>Range and Accuracy</td><td colspan="2">Range: -10°C to 120°C Accuracy ±1°C,</td></tr>
<tr><td>Onboard Data Storage</td><td>EEPROM</td><td colspan="2">uSD Card: every 1 second, logged over last 60 days
EEPROM: every 90 seconds, logged over last 31 days</td></tr>
<tr><td rowspan="3">Over Voltage Protection</td><td>Threshold, Protection Trigger Delay</td><td colspan="2">Configurable</td></tr>
<tr><td>Fault Response</td><td colspan="2">Charge MOSFET: OFF, Discharge MOSFET: ON</td></tr>
<tr><td>Fault Release</td><td colspan="2">Through Hysteresis, Automatic</td></tr>
<tr><td rowspan="3">Under Voltage Protection</td><td>Threshold, Protection Trigger Delay</td><td colspan="2">Configurable</td></tr>
<tr><td>Fault Response</td><td colspan="2">Discharge MOSFET: OFF, Charge MOSFET: ON</td></tr>
<tr><td>Fault Release</td><td>Configurable Hysteresis</td><td>No load removal required. Charging current is allowed.</td></tr>
<tr><td rowspan="3">Over Current in Discharge Protection Level-1</td><td>Threshold, Protection Trigger Delay</td><td colspan="2">Configurable to allow up to 120A discharge for up to 30 seconds</td></tr>
<tr><td>Fault Response</td><td colspan="2">Discharge MOSFET: OFF, Charge MOSFET: ON</td></tr>
<tr><td>Fault Release</td><td colspan="2">Auto recovery after no discharge current for 90 secs</td></tr>
<tr><td rowspan="3">Over Current in Discharge Protection Level-2</td><td>Threshold, Protection Trigger Delay</td><td colspan="2">Configurable to instantaneously cut off at 140A discharge current in 100 mSec</td></tr>
<tr><td>Fault Response</td><td colspan="2">Discharge MOSFET: OFF, Charge MOSFET: ON</td></tr>
<tr><td>Fault Release</td><td colspan="2">Auto recovery after no discharge current for 90 secs</td></tr>
</table>

Note: Specifications are indicative only. Final specifications and behaviour including continuous discharge/charge current, balancing current, protection thresholds, etc. may vary depending upon BMS version, BMS mounting scheme, PDU heat dissipation, programmed parameters and application logic.

Specifications		Remarks
Over Current in Discharge Latch Protection	Threshold	5 consecutive over discharge current faults
	Fault Response	Discharge MOSFET: OFF, Charge MOSFET: ON
	Fault Release	Load Removal or Auto release
Over Current in Charge Protection Level-1	Threshold, Protection Trigger Delay	Configurable to allow up to **120A** charge current for up to 30 seconds
	Fault Response	Discharge MOSFET: ON, Charge MOSFET: OFF
	Fault Release	Auto recovery after no charge current for 90 secs
Over Current in Charge Protection Level-2	Threshold, Protection Trigger Delay	Configurable to instantaneously cut off at **140A** charge current in 100 mSec
	Fault Response	Charge MOSFET: OFF, Discharge MOSFET: ON
	Fault Release	Auto recovery after no charge current for 90 secs
Short Circuit in Discharge Protection	Protection Trigger Delay	~30 uSec
	Fault Response	Discharge MOSFET: OFF, Charge MOSFET: OFF
	Fault Release	Auto recovery after 5 seconds
Short Circuit in Discharge Latch Protection	Threshold	2 consecutive short circuit faults
	Fault Response	Discharge MOSFET: OFF, Charge MOSFET: OFF
	Fault Release	Load Removal
Over Temperature in Discharge Protection	Threshold, Protection Trigger Delay	Configurable
	Fault Response	Discharge MOSFET: OFF, Charge MOSFET: ON
	Fault Release	Auto recovery below configurable temperature
Under Temperature in Discharge Protection	Threshold, Protection Trigger Delay	Configurable
	Fault Response	Discharge MOSFET: OFF, Charge MOSFET: ON
	Fault Release	Auto recovery above configurable temperature
Over Temperature in Charge Protection	Threshold, Protection Trigger Delay	Configurable
	Fault Response	Charge MOSFET: OFF, Discharge MOSFET: ON
	Fault Release	Auto recovery below configurable temperature
Under Temperature in Charge Protection	Threshold, Protection Trigger Delay	Configurable
	Fault Response	Charge MOSFET: OFF, Discharge MOSFET: ON
	Fault Release	Auto recovery above configurable temperature
Buzzer Output	Alert on Permanent Failure	Thermal Runaway, MOSFET Failure, Cell Open Wire, Short Circuit Latch, Over Current in Discharge Latch, Insulation Failure

Pin Diagram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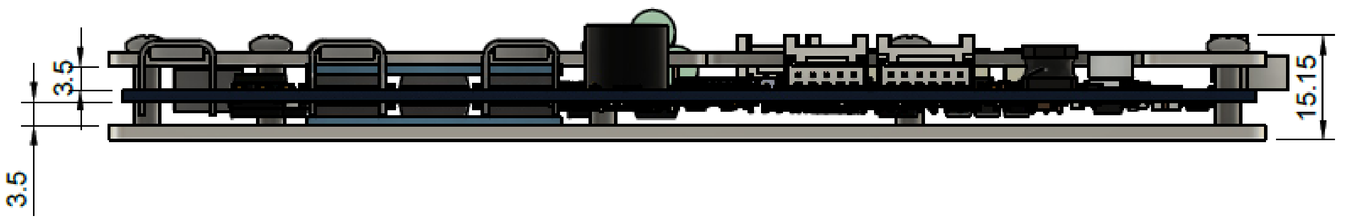
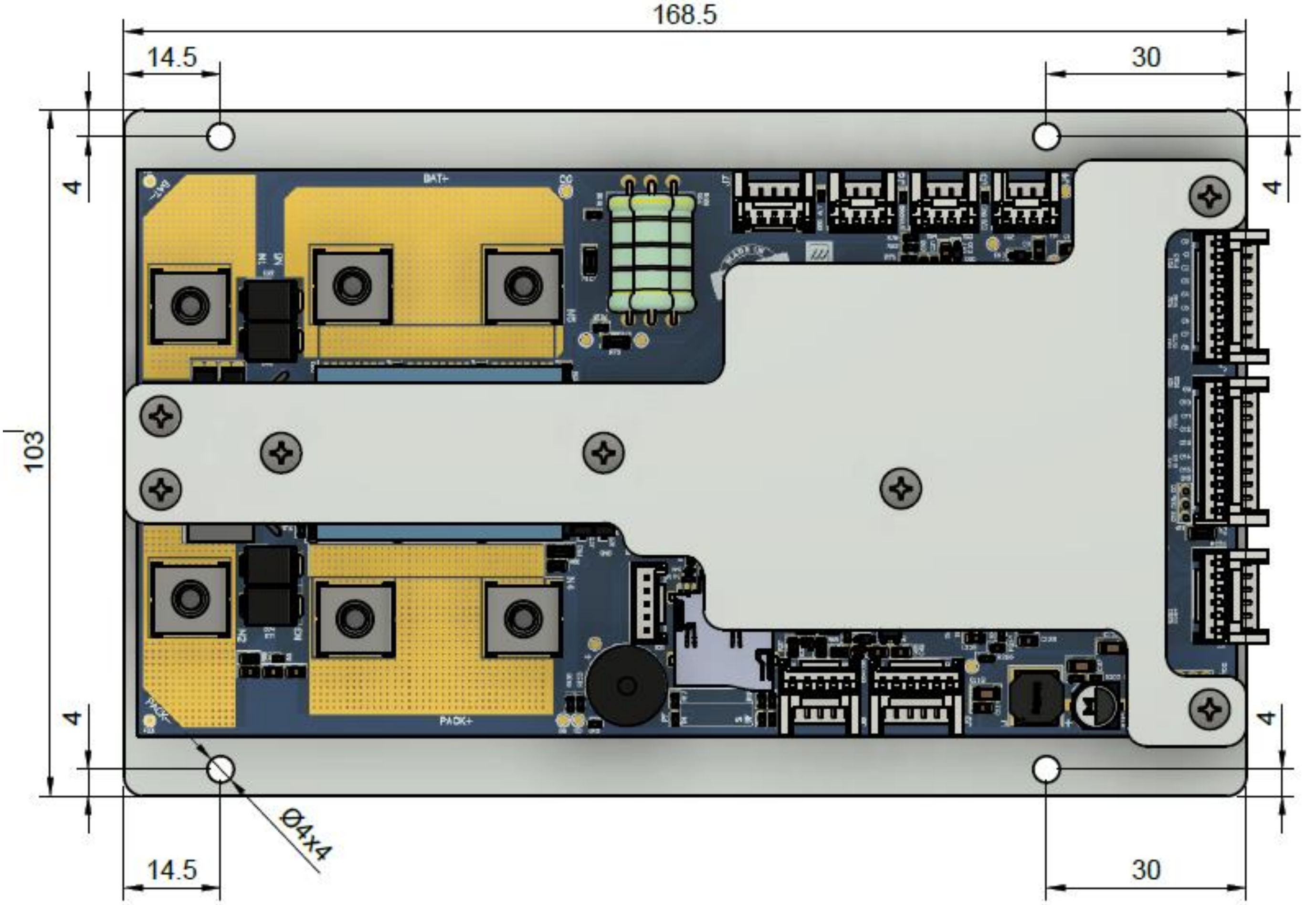
C17
C18
C19
C20
C20S
TS1
TS2
TS3
TS4
TS5
TS6
GPIO1
RX
TX
GND
5V
GPI1
GND
GPI2
GND
CANH
CANL
GPIO1
RX
TX
GND
5V
BAT+
BAT-
PACK+
PACK-

Mechanical Dimensions





All dimensions are in mm